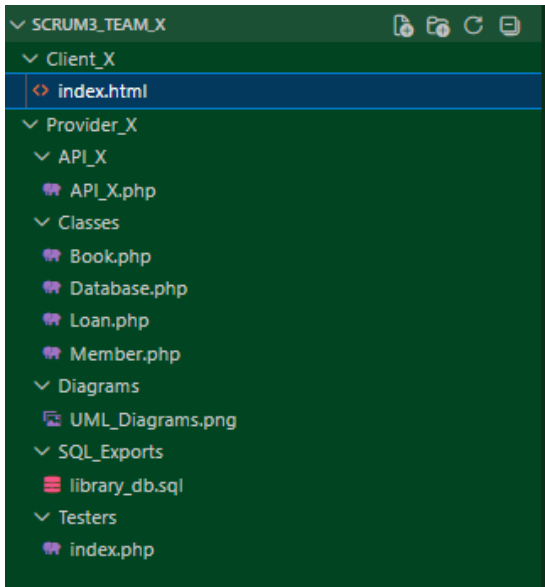


Team: Ezra, Malik, Ethan
Instructor: Huda Judeh
Class: Class 266 - O
Date: March, 28, 2026
Assignment: Scrum 3 - Library Management Web Service IO Captures

Screen 1 - Folder Structure (Index of /Scrum3_Team_X)

Root folder containing Client_X/ and Provider_X/ confirming correct structure.



Screen 2 - Testers/index.php - Loans JSON Output + Books HTML Table

All 8 books returned in JSON format from API_X.php via fetch().

Loans - getAll JSON Response:

Raw JSON:

```
[{"id":1,"book_id":1,"member_id":1,"loan_date":"2025-01-01","return_date":"2025-01-10"}, {"id":2,"book_id":2,"member_id":2,"loan_date":"2025-01-05","return_date":"2025-01-15"}, {"id":3,"book_id":3,"member_id":3,"loan_date":"2025-01-10","return_date":"2025-01-20"}, {"id":4,"book_id":4,"member_id":4,"loan_date":"2025-01-12","return_date":"2025-01-22"}, {"id":5,"book_id":5,"member_id":5,"loan_date":"2025-01-15","return_date":"2025-01-25"}, {"id":6,"book_id":6,"member_id":6,"loan_date":"2025-01-18","return_date":"2025-01-28"}, {"id":7,"book_id":7,"member_id":7,"loan_date":"2025-01-20","return_date":"2025-01-30"}, {"id":8,"book_id":8,"member_id":8,"loan_date":"2026-03-28","return_date":"0000-00-00"}]
```

Decoded Response:

ID: 1 | Book ID: 1 | Member ID: 1 | Loan Date: 2025-01-01 | Return Date: 2025-01-10

ID: 2 | Book ID: 2 | Member ID: 2 | Loan Date: 2025-01-05 | Return Date: 2025-01-15

ID: 3 | Book ID: 3 | Member ID: 3 | Loan Date: 2025-01-10 | Return Date: 2025-01-20

ID: 4 | Book ID: 4 | Member ID: 4 | Loan Date: 2025-01-12 | Return Date: 2025-01-22

ID: 5 | Book ID: 5 | Member ID: 5 | Loan Date: 2025-01-15 | Return Date: 2025-01-25

ID: 6 | Book ID: 6 | Member ID: 6 | Loan Date: 2025-01-18 | Return Date: 2025-01-28

ID: 7 | Book ID: 7 | Member ID: 7 | Loan Date: 2025-01-20 | Return Date: 2025-01-30

ID: 8 | Book ID: 8 | Member ID: 8 | Loan Date: 2026-03-28 | Return Date: 0000-00-00

Books

ID	Title	Author	Genre	Year
1	Testing	Testing Genre	Testing Genre 1	1901
2	1984	George Orwell	Dystopian	1949
3	To Kill a Mockingbird	Harper Lee	Fiction	1960
4	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
5	Moby Dick	Herman Melville	Adventure	1851
6	Harry Potter	J.K. Rowling	Fantasy	1997
7	The Alchemist	Paulo Coelho	Fiction	1988
8	Test	Testing	Testing Genre	1900

Screen 3 - Client_X Books Tab - Get Book by ID, Add Book, Update Book

Get Book ID 1 returns The Hobbit. Add Book returns {"success": true}. Update Book returns {"success": true}.

```
{
  "author": "Paulo Coelho",
  "genre": "Fiction",
  "year": 1988
}
```

Get Book by ID

```
{
  "id": 1,
  "title": "The Hobbit",
  "author": "J.R.R. Tolkien",
  "genre": "Fantasy",
  "year": 1937
}
```

Add Book

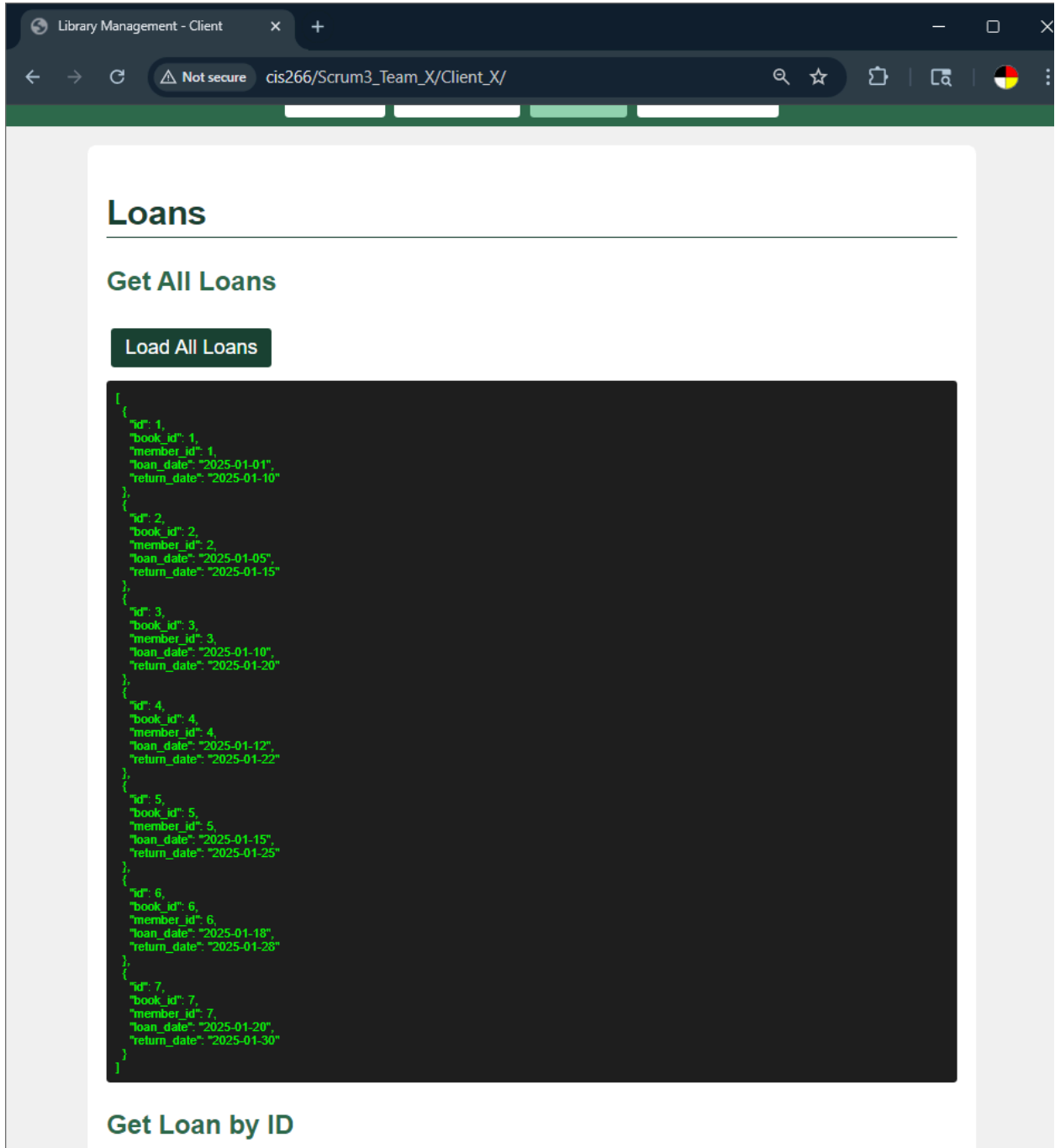
```
{
  "success": true
}
```

Update Book

```
{
  "success": true
}
```

Screen 4 - Client_X Loans Tab - Load All Loans

All 7 members returned in JSON. Get Member ID 1 returns Alice Johnson.



The screenshot shows a web browser window with the address bar displaying 'Library Management - Client' and the URL 'cis266/Scrum3_Team_X/Client_X/'. The page title is 'Loans'. Below the title, there is a section titled 'Get All Loans' with a button labeled 'Load All Loans'. The button is clicked, and a JSON array of 7 loan records is displayed in a dark-themed code editor. The JSON data is as follows:

```
[
  {
    "id": 1,
    "book_id": 1,
    "member_id": 1,
    "loan_date": "2025-01-01",
    "return_date": "2025-01-10"
  },
  {
    "id": 2,
    "book_id": 2,
    "member_id": 2,
    "loan_date": "2025-01-05",
    "return_date": "2025-01-15"
  },
  {
    "id": 3,
    "book_id": 3,
    "member_id": 3,
    "loan_date": "2025-01-10",
    "return_date": "2025-01-20"
  },
  {
    "id": 4,
    "book_id": 4,
    "member_id": 4,
    "loan_date": "2025-01-12",
    "return_date": "2025-01-22"
  },
  {
    "id": 5,
    "book_id": 5,
    "member_id": 5,
    "loan_date": "2025-01-15",
    "return_date": "2025-01-25"
  },
  {
    "id": 6,
    "book_id": 6,
    "member_id": 6,
    "loan_date": "2025-01-18",
    "return_date": "2025-01-28"
  },
  {
    "id": 7,
    "book_id": 7,
    "member_id": 7,
    "loan_date": "2025-01-20",
    "return_date": "2025-01-30"
  }
]
```

Below the JSON data, there is a section titled 'Get Loan by ID'.

Screen 5 - Client_X Members Tab - Add Member + Delete Member

Add Member returns {"success": true}. Delete Member ID 8 returns {"success": true}.

Added

Library Management - Client

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Get All Members

Load All Members

```
{
  "id": 1,
  "name": "Alice Johnson",
  "email": "alice@example.com",
  "phone": "12345678",
  "join_date": "2024-01-10"
},
{
  "id": 2,
  "name": "Bob Smith",
  "email": "bob@example.com",
  "phone": "87654321",
  "join_date": "2024-02-15"
},
{
  "id": 3,
  "name": "Charlie Brown",
  "email": "charlie@example.com",
  "phone": "11223344",
  "join_date": "2024-03-20"
},
{
  "id": 4,
  "name": "Diana Prince",
  "email": "diana@example.com",
  "phone": "22334455",
  "join_date": "2024-04-05"
},
{
  "id": 5,
  "name": "Ethan Hunt",
  "email": "ethan@example.com",
  "phone": "33445566",
  "join_date": "2024-05-12"
},
{
  "id": 6,
  "name": "Fiona Green",
  "email": "fiona@example.com",
  "phone": "44556677",
  "join_date": "2024-06-18"
},
{
  "id": 7,
  "name": "George King",
  "email": "george@example.com",
  "phone": "55667788",
  "join_date": "2024-07-22"
},
{
  "id": 8,
  "name": "David Martinez",
  "email": "DMartinez@ERunner.com",
  "phone": "099-999-9999",
  "join_date": "2077-08-18"
}
}
```

Get Member by ID

Get Member

JSON will appear here...

Deleted

Library Management - Client

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```
{
  "id": 7,
  "name": "George King",
  "email": "george@example.com",
  "phone": "55067788",
  "join_date": "2024-07-22"
},
{
  "id": 8,
  "name": "David Martinez",
  "email": "DMartinez@ERunner.com",
  "phone": "999-999-9999",
  "join_date": "2077-08-18"
}
}
```

Get Member by ID

JSON will appear here...

Add Member

{
 "success": true
}

Update Member

JSON will appear here...

Delete Member

{
 "success": true
}

Screen 6 - Client_X Loans Tab - Load All Loans + Get by ID

All 7 loans returned in JSON. Get Loan ID 1 returns correctly.

Library Management - Client

Not secure cis266/Scrum3_Team_X/Client_X/

Get All Loans

Load All Loans

```
{
  {
    "id": 1,
    "book_id": 1,
    "member_id": 1,
    "loan_date": "2025-01-01",
    "return_date": "2025-01-10"
  },
  {
    "id": 2,
    "book_id": 2,
    "member_id": 2,
    "loan_date": "2025-01-05",
    "return_date": "2025-01-15"
  },
  {
    "id": 3,
    "book_id": 3,
    "member_id": 3,
    "loan_date": "2025-01-10",
    "return_date": "2025-01-20"
  },
  {
    "id": 4,
    "book_id": 4,
    "member_id": 4,
    "loan_date": "2025-01-12",
    "return_date": "2025-01-22"
  },
  {
    "id": 5,
    "book_id": 5,
    "member_id": 5,
    "loan_date": "2025-01-15",
    "return_date": "2025-01-25"
  },
  {
    "id": 6,
    "book_id": 6,
    "member_id": 6,
    "loan_date": "2025-01-18",
    "return_date": "2025-01-28"
  },
  {
    "id": 7,
    "book_id": 7,
    "member_id": 7,
    "loan_date": "2025-01-20",
    "return_date": "2025-01-30"
  }
}
```

Get Loan by ID

1 Get Loan

```
{
  "id": 1,
  "book_id": 1,
  "member_id": 1,
  "loan_date": "2025-01-01",
  "return_date": "2025-01-10"
}
```

Add Loan

Screen 7 - Client_X Loans Tab - Add Loan + Delete Loan

Add Loan returns {"success": true}. Delete Loan returns {"success": true}.

Library Management - Client

Not secure cis266/Scrum3_Team_X/Client_X/

```
{
  "return_date": "2025-01-28"
},
{
  "id": 7,
  "book_id": 7,
  "member_id": 7,
  "loan_date": "2025-01-20",
  "return_date": "2025-01-30"
}
}
```

Get Loan by ID

Get Loan

```
{
  "id": 7,
  "book_id": 7,
  "member_id": 7,
  "loan_date": "2025-01-20",
  "return_date": "2025-01-30"
}
```

Add Loan

Add Loan

```
{
  "success": true
}
```

Update Loan

Update Loan

JSON will appear here...

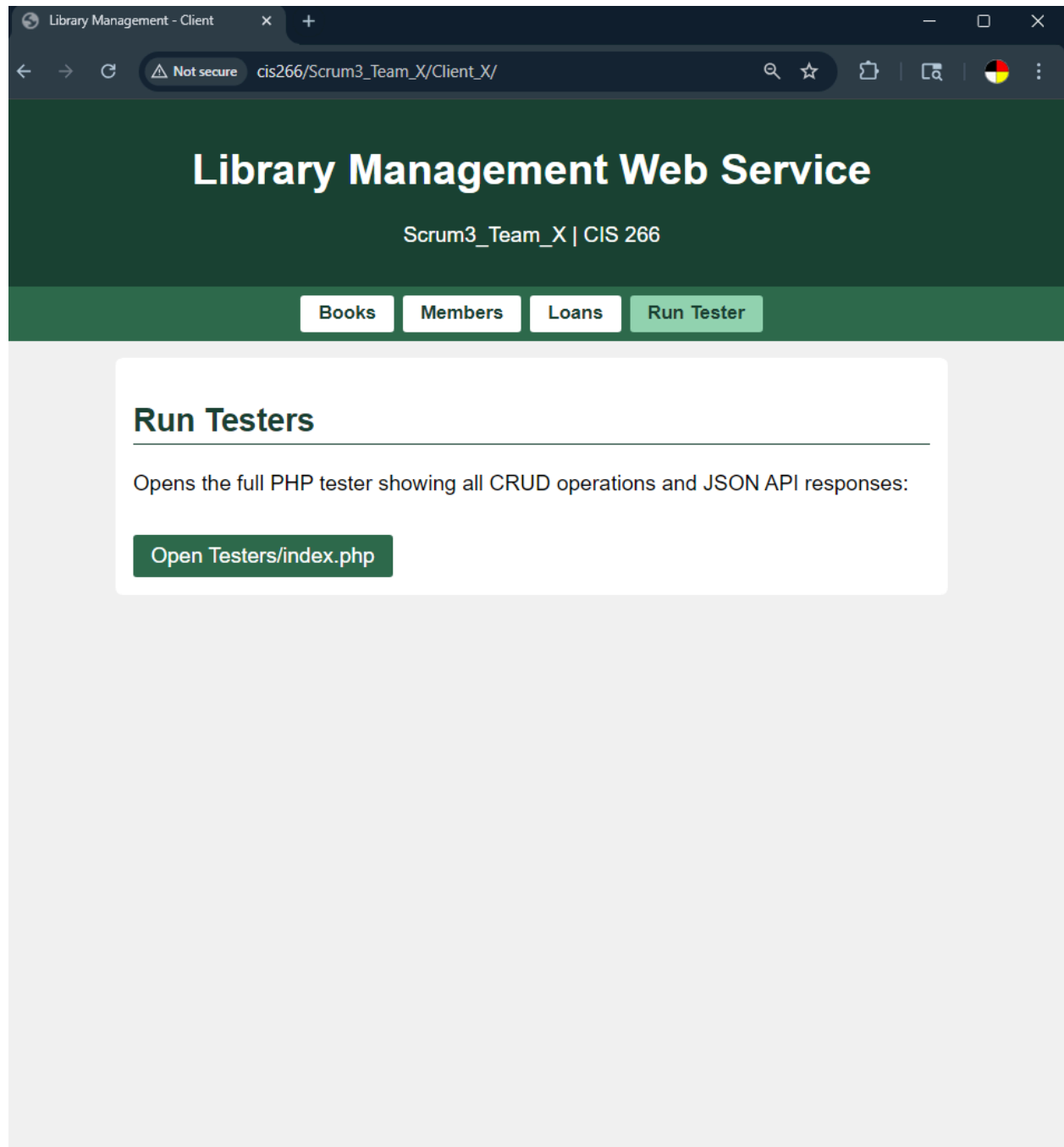
Delete Loan

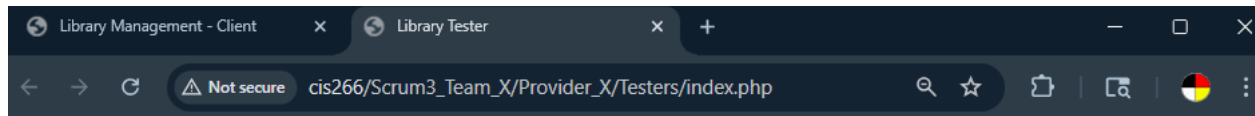
Delete Loan

```
{
  "success": true
}
```

Screen 8 - Client_X Run Tester Tab

Button linking to Testers/index.php, confirming navigation works.





Library Tester - Scrum3_Team_X

API JSON Output Tester

Books - getAll JSON Response:

Raw JSON:

```
[{"id":1,"title":"Testing ","author":"Testing Genre ","genre":"Testing Genre 1","year":1901},{"id":2,"title":"1984","author":"George Orwell","genre":"Dystopian","year":1949},{"id":3,"title":"To Kill a Mockingbird","author":"Harper Lee","genre":"Fiction","year":1960},{"id":4,"title":"The Great Gatsby","author":"F. Scott Fitzgerald","genre":"Classic","year":1925},{"id":5,"title":"Moby Dick","author":"Herman Melville","genre":"Adventure","year":1851},{"id":6,"title":"Harry Potter","author":"J.K. Rowling","genre":"Fantasy","year":1997},{"id":7,"title":"The Alchemist","author":"Paulo Coelho","genre":"Fiction","year":1988}]
```

Decoded Response:

ID: 1 | **Title:** Testing | **Author:** Testing Genre | **Genre:** Testing Genre 1 | **Year:** 1901

ID: 2 | **Title:** 1984 | **Author:** George Orwell | **Genre:** Dystopian | **Year:** 1949

ID: 3 | **Title:** To Kill a Mockingbird | **Author:** Harper Lee | **Genre:** Fiction | **Year:** 1960

ID: 4 | **Title:** The Great Gatsby | **Author:** F. Scott Fitzgerald | **Genre:** Classic | **Year:** 1925

ID: 5 | **Title:** Moby Dick | **Author:** Herman Melville | **Genre:** Adventure | **Year:** 1851

ID: 6 | **Title:** Harry Potter | **Author:** J.K. Rowling | **Genre:** Fantasy | **Year:** 1997

ID: 7 | **Title:** The Alchemist | **Author:** Paulo Coelho | **Genre:** Fiction | **Year:** 1988

Library Management - Client

Library Tester

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ISBN: 9780316066525 | Title: The Alchemist | Author: Paulo Coelho | Genre: Fiction | Year: 1988

ID: 7 | Title: The Alchemist | Author: Paulo Coelho | Genre: Fiction | Year: 1988

Members - getAll JSON Response:

Raw JSON:

```
[{"id":1,"name":"Alice Johnson","email":"alice@example.com","phone":"12345678","join_date":"2024-01-10"}, {"id":2,"name":"Bob Smith","email":"bob@example.com","phone":"87654321","join_date":"2024-02-15"}, {"id":3,"name":"Charlie Brown","email":"charlie@example.com","phone":"11223344","join_date":"2024-03-20"}, {"id":4,"name":"Diana Prince","email":"diana@example.com","phone":"22334455","join_date":"2024-04-05"}, {"id":5,"name":"Ethan Hunt","email":"ethan@example.com","phone":"33445566","join_date":"2024-05-12"}, {"id":6,"name":"Fiona Green","email":"fiona@example.com","phone":"44556677","join_date":"2024-06-18"}, {"id":7,"name":"George King","email":"george@example.com","phone":"55667788","join_date":"2024-07-22"}]
```

Decoded Response:

ID: 1 | Name: Alice Johnson | Email: alice@example.com | Phone: 12345678 | Join Date: 2024-01-10

ID: 2 | Name: Bob Smith | Email: bob@example.com | Phone: 87654321 | Join Date: 2024-02-15

ID: 3 | Name: Charlie Brown | Email: charlie@example.com | Phone: 11223344 | Join Date: 2024-03-20

ID: 4 | Name: Diana Prince | Email: diana@example.com | Phone: 22334455 | Join Date: 2024-04-05

ID: 5 | Name: Ethan Hunt | Email: ethan@example.com | Phone: 33445566 | Join Date: 2024-05-12

ID: 6 | Name: Fiona Green | Email: fiona@example.com | Phone: 44556677 | Join Date: 2024-06-18

ID: 7 | Name: George King | Email: george@example.com | Phone: 55667788 | Join Date: 2024-07-22

Loans - getAll JSON Response:

Raw JSON:

```
[{"id":1,"book_id":1,"member_id":1,"loan_date":"2025-01-01","return_date":"2025-01-10"}, {"id":2,"book_id":2,"member_id":2,"loan_date":"2025-01-05","return_date":"2025-01-15"}, {"id":3,"book_id":3,"member_id":3,"loan_date":"2025-01-10","return_date":"2025-01-20"}, {"id":4,"book_id":4,"member_id":4,"loan_date":"2025-01-12","return_date":"2025-01-22"}, {"id":5,"book_id":5,"member_id":5,"loan_date":"2025-01-15","return_date":"2025-01-25"}, {"id":6,"book_id":6,"member_id":6,"loan_date":"2025-01-18","return_date":"2025-01-28"}, {"id":10,"book_id":7,"member_id":7,"loan_date":"2025-01-20","return_date":"2025-01-30"}]
```

Decoded Response:

ID: 1 | Book ID: 1 | Member ID: 1 | Loan Date: 2025-01-01 | Return Date: 2025-01-10

ID: 2 | Book ID: 2 | Member ID: 2 | Loan Date: 2025-01-05 | Return Date: 2025-01-15

ID: 3 | Book ID: 3 | Member ID: 3 | Loan Date: 2025-01-10 | Return Date: 2025-01-20

ID: 4 | Book ID: 4 | Member ID: 4 | Loan Date: 2025-01-12 | Return Date: 2025-01-22

ID: 5 | Book ID: 5 | Member ID: 5 | Loan Date: 2025-01-15 | Return Date: 2025-01-25

ID: 6 | Book ID: 6 | Member ID: 6 | Loan Date: 2025-01-18 | Return Date: 2025-01-28

ID: 10 | Book ID: 7 | Member ID: 7 | Loan Date: 2025-01-20 | Return Date: 2025-01-30

Books

Title

Author

Genre

Year

Add Book

Book ID

New Title

New Author

New Genre

New Year

Update Book

Book ID

Delete Book

ID	Title	Author	Genre	Year
1	Testing	Testing Genre	Testing Genre 1	1901
2	1984	George Orwell	Dystopian	1949
3	To Kill a Mockingbird	Harper Lee	Fiction	1960
4	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
5	Moby Dick	Herman Melville	Adventure	1851
6	Harry Potter	J.K. Rowling	Fantasy	1997
7	The Alchemist	Paulo Coelho	Fiction	1988

Members

Name

Email

Phone

Add Member

Member ID

New Name

New Email

New Phone

Update Member

Member ID

Delete Member

ID	Name	Email	Phone	Join Date
1	Alice Johnson	alice@example.com	12345678	2024-01-10
2	Bob Smith	bob@example.com	87654321	2024-02-15
3	Charlie Brown	charlie@example.com	11223344	2024-03-20
4	Diana Prince	diana@example.com	22334455	2024-04-05
5	Ethan Hunt	ethan@example.com	33445566	2024-05-12
6	Fiona Green	fiona@example.com	44556677	2024-06-18
7	George King	george@example.com	55667788	2024-07-22

Loans

Book ID

Member ID

Add Loan

Loan ID

Book ID

Member ID

Update Loan

Members

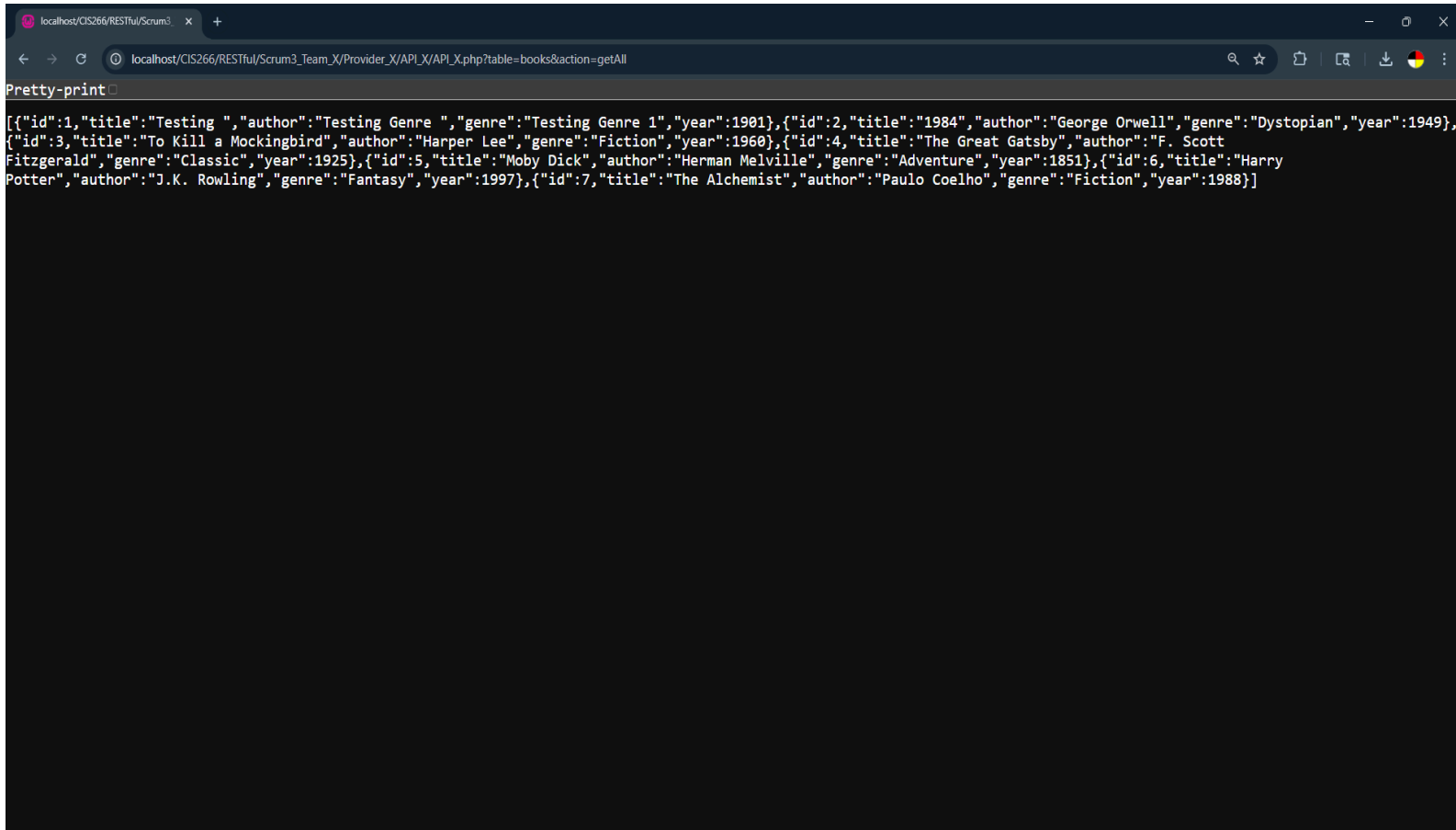
ID	Name	Email	Phone	Join Date
1	Alice Johnson	alice@example.com	12345678	2024-01-10
2	Bob Smith	bob@example.com	87654321	2024-02-15
3	Charlie Brown	charlie@example.com	11223344	2024-03-20
4	Diana Prince	diana@example.com	22334455	2024-04-05
5	Ethan Hunt	ethan@example.com	33445566	2024-05-12
6	Fiona Green	fiona@example.com	44556677	2024-06-18
7	George King	george@example.com	55667788	2024-07-22

Loans

ID	Book ID	Member ID	Loan Date	Return Date
1	1	1	2025-01-01	2025-01-10
2	2	2	2025-01-05	2025-01-15
3	3	3	2025-01-10	2025-01-20
4	4	4	2025-01-12	2025-01-22
5	5	5	2025-01-15	2025-01-25
6	6	6	2025-01-18	2025-01-28
10	7	7	2025-01-20	2025-01-30

Screen 9 - Testers/index.php - Full JSON Output + HTML Tables

All 3 tables (Books, Members, Loans) displaying raw JSON and decoded responses. All 3 HTML CRUD tables displaying all records. "Loan deleted!" confirm message visible.



The screenshot shows a web browser window with a REST client interface. The address bar displays the URL: `localhost/CIS266/RESTful/Scrum3_Team_X/Provider_X/API_X/API_X.php?table=books&action=getAll`. The response is displayed in a dark-themed editor with a "Pretty-print" button. The JSON response is as follows:

```
[{"id":1,"title":"Testing ","author":"Testing Genre ","genre":"Testing Genre 1","year":1901}, {"id":2,"title":"1984","author":"George Orwell","genre":"Dystopian","year":1949}, {"id":3,"title":"To Kill a Mockingbird","author":"Harper Lee","genre":"Fiction","year":1960}, {"id":4,"title":"The Great Gatsby","author":"F. Scott Fitzgerald","genre":"Classic","year":1925}, {"id":5,"title":"Moby Dick","author":"Herman Melville","genre":"Adventure","year":1851}, {"id":6,"title":"Harry Potter","author":"J.K. Rowling","genre":"Fantasy","year":1997}, {"id":7,"title":"The Alchemist","author":"Paulo Coelho","genre":"Fiction","year":1988}]
```